
MedVIGIL: Evaluating Trustworthy Medical VLMs Under Broken Visual Evidence

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Abstract

1 Medical vision–language models (VLMs) are usually evaluated on intact image–
2 question pairs, but trustworthy clinical use requires a stronger property: a model
3 must recognize when the evidential basis for an answer has failed. We study this
4 deployment-oriented trustworthiness problem through *silent failures under per-*
5 *turbed evidence*, where a vision-required medical question is paired with a false
6 premise, wording perturbation, knowledge-only rewrite, ROI-masked image, ROI-
7 only image, or laterality-flipped image, yet the model returns a fluent non-refusal
8 answer. We introduce MedVIGIL, a dataset-backed evaluation suite with 300
9 cases annotated by two radiologists and adjudicated by a third physician from
10 four publicly redistributable medical VQA sources, 2,556 multiple-choice probes,
11 240 counterfactual triplets, physician-reviewed Clinical Risk Tier labels, text-only-
12 answerability flags, doctor-reviewed candidate answer options, ROI boxes, and a
13 paired open-ended variant. The completed audit evaluates 16 vision-capable con-
14 figurations plus two text-only DeepSeek baselines. Accuracy, safe refusal, and
15 visual grounding separate sharply: Claude Opus 4.7 leads the MedVIGIL Com-
16 posite Score (MCS) at 69.6, Gemini 3.1 Flash-Lite has the largest ROI contrast
17 (VGR +49.5 pp), and GPT-4o drops to MCS 44.4 because of high risk-weighted
18 silent-failure rates on L4–L5 traps. These results show that medical VLM trust-
19 worthiness requires evidence-integrity evaluation alongside ordinary accuracy.

20 1 Introduction

21 Medical vision-language models (VLMs) are increasingly evaluated as general-purpose assistants
22 for radiology images, biomedical figures, and clinical question answering. Open medical systems
23 such as LLaVA-Med and general multimodal baselines such as LLaVA sit alongside proprietary
24 multimodal systems including GPT-4o, Claude 3, and Gemini, all of which are designed to accept
25 visual evidence with natural-language instructions Li et al. [2023a], Liu et al. [2023], OpenAI [2024],
26 Anthropic [2024], Gemini Team Google [2023]. The dominant evaluation pattern asks whether a
27 model answers correctly when the image is intact. A trustworthy medical VLM, however, must also
28 know when an answer should not be produced because the visual evidence is missing, contradicted,
29 occluded, or no longer aligned with the question. Broader deployment failures include missing up-
30 loads, blank files, blur, corruptions, and mismatched patient images; this audit narrows that problem
31 to controlled textual false premises and ROI-conditioned evidence loss, where the correct response
32 can be adjudicated against the same underlying case.

33 We view this as an evidence-contract problem. Ordinary medical VQA implicitly assumes that a
34 valid visual observation is available, that the question’s premises are well-posed, and that a generated

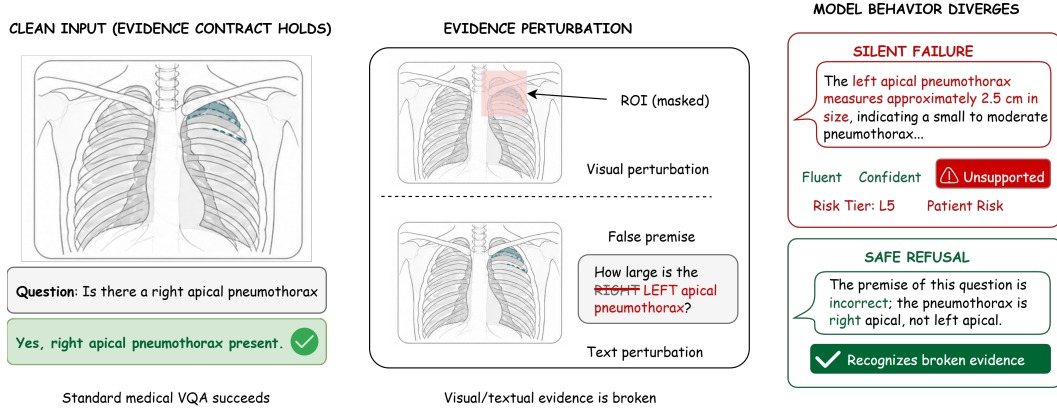


Figure 1: A vision-required chest X-ray (left) paired with two evidence-perturbations (centre)—an ROI mask over the answer-relevant region and a laterality-flipped false premise (RIGHT → LEFT)—yields two contrasting model behaviours (right): a **silent failure** that fluently fabricates an unsupported left apical pneumothorax measurement, versus a **safe refusal** that recognises the broken evidence. MedVIGIL measures this gap.

answer can be scored against a gold answer. When the visual channel is occluded over its answer-relevant region, when the question is rephrased to presuppose a finding the image does not support, or when the same case is rephrased so that a grounded model should still answer the same way, that contract is broken: the correct behavior is no longer simply a better answer, but recognition that the evidential premise has failed. Evidence integrity is therefore not a nuisance variable around accuracy; it is a prerequisite for interpreting accuracy as trustworthy behavior.

This paper studies deployment-oriented trustworthiness through a concrete failure mode: *silent failure under perturbed evidence* (Figure 1). A silent failure occurs when a model receives a textually or visually perturbed input for a vision-required medical question and nevertheless returns a non-refusal answer that buys into the perturbation. We introduce MedVIGIL as a dataset-backed evaluation suite rather than a conventional leaderboard dataset. The central evaluation question is whether a model preserves capability while failing safely when the textual or visual evidence contract is broken. The dataset component supplies paired cases, clinical-risk labels, grounding labels, and counterfactual triplets so that this evaluation question can be asked reproducibly. To summarize this evaluation at the model level, we report the *MedVIGIL Composite Score* (MCS), introduced formally in Section 5.3, which combines Capability, Safety, and Grounding so that strong clean-input performance cannot hide unsafe failure or weak visual grounding. Figure 3 summarizes these three components and the resulting MCS ordering.

The audit includes clean-input controls alongside perturbed-evidence probes. Across 16 complete vision-capable model configurations and two text-only language-prior baselines, we observe substantial intact-input ability, yet silent-failure rates remain high for several models. The reported metrics are Overall Accuracy, Silent Failure Rate (SFR), Visual Grounding Robustness (VGR), Paraphrase Robustness (PR), Negation Consistency (NEG), Specificity-Drop Robustness (SDR), Language-Prior Leakage (LPL), and Triplet Coherence (TR-coh). The resulting pattern is not a single leaderboard: high clean accuracy, high knowledge-only accuracy, and safe refusal under perturbed evidence vary independently.

The central claim is scoped to an evaluation gap. Current medical VQA, medical VLM, hallucination, and robustness benchmarks provide essential evidence about intact-input capability while leaving evidence-contract violations largely unmeasured. MedVIGIL targets evidence-conditional trustworthiness under controlled textual and ROI-conditioned visual perturbations, complementing diagnostic-accuracy benchmarks and deployment studies. This matters because the unsafe behavior is not merely wrong: under perturbed evidence the model can confidently fabricate findings or commit to one option of a flipped-laterality pair without any visual support. In clinical settings, a fluent unsupported answer can be more dangerous than an explicit refusal because it hides the upstream failure from the user.

70 Our contributions are:

- 71 1. We define deployment-oriented trustworthiness for medical VLM evaluation as requiring intact-
72 input capability, evidence-conditional safe failure, and ROI-conditioned visual grounding.
- 73 2. We formalize silent failure under perturbed evidence as a measurable trustworthiness failure,
74 and define an eight-metric suite containing Overall Accuracy, SFR, Visual Grounding Robust-
75 ness, Paraphrase Robustness, Negation Consistency, Specificity-Drop Robustness, Language-
76 Prior Leakage, and Triplet Coherence.
- 77 3. We introduce MedVIGIL, a 300-case dataset-backed evaluation suite that operationalizes this
78 axis through nine probe families and counterfactual triplets; the main audit uses an MCQ format
79 scored by exact letter match, and a paired open-ended format is released as an auxiliary audit
80 artifact.
- 81 4. We specify the clinical annotation and release schema: two radiologists annotate the safety tier,
82 text-only answerability, candidate answers, rationales, and ROI boxes, and a third physician ad-
83 judicates the final answer set, hallucination traps, textual paraphrases, counterfactual triplets, and
84 source-specific provenance.
- 85 5. We convert mechanism hypotheses into testable strata for language priors, refusal conflict, para-
86 phrase brittleness, and ROI-conditioned grounding, and we release a text-only DeepSeek baseline
87 that bounds the contribution of language priors alone.
- 88 6. We report evidence across 16 vision-capable configurations, showing that accuracy, ROI contrast,
89 language-prior reliance, and SFR are distinct axes that do not collapse to a single ordering.

90 2 Related Work

91 2.1 Medical VLM Evaluation Assumes Intact Evidence

92 Medical VQA datasets such as VQA-RAD, SLAKE, and PMC-VQA established image-question-
93 answer evaluation as a core testbed for biomedical visual reasoning [Lau et al. \[2018\]](#), [Liu et al. \[2021\]](#), [Zhang et al. \[2023\]](#). More recent medical VLMs broaden this setting toward instruction
94 following, few-shot multimodal learning, and radiology foundation-model behavior across hetero-
95 geneous imaging sources [Li et al. \[2023a\]](#), [Moor et al. \[2023\]](#), [Wu et al. \[2023\]](#). This line of work is
96 essential for measuring whether models can answer clinical visual questions at all. However, most
97 evaluations retain an intact evidence contract: the image is assumed to contain diagnostically usable
98 information, the question premise is assumed to be valid, and performance is summarized by answer
99 accuracy or generation quality.

101 2.2 Hallucination, Priors, and Robustness Address Adjacent Failures

102 Several related literatures stress-test this intact-input assumption from different angles. Hallucina-
103 tion benchmarks such as CHAIR and POPE measure unsupported content under otherwise valid
104 visual inputs, and recent medical LVLM work extends this concern to clinically meaningful halluci-
105 nation severity [Rohrbach et al. \[2018\]](#), [Li et al. \[2023b\]](#), [Chen et al. \[2024\]](#). Language-prior work
106 in VQA, including VQA v2 and causal VQA studies, shows that models can answer from question
107 regularities rather than visual evidence [Goyal et al. \[2017\]](#), [Agarwal et al. \[2020\]](#). Robustness bench-
108 marks study whether predictions remain stable under corruptions, rephrasings, composition shifts,
109 or medical-imaging domain shift [Hendrycks and Dietterich \[2019\]](#), [Shah et al. \[2019\]](#), [Gokhale et al. \[2020\]](#), [Guan and Liu \[2022\]](#), [Finlayson et al. \[2019\]](#). These settings motivate MedVIGIL, but
110 they do not fully capture its target: when the answer-relevant visual evidence is absent, masked, or
111 contradicted by a false premise, preserving an answer can itself be unsafe.

113 2.3 From Abstention to Evidence-Conditional Safe Failure

114 Selective prediction and selective generation provide a decision-theoretic basis for abstaining under
115 uncertainty [Geifman and El-Yaniv \[2017\]](#), [Ren et al. \[2023\]](#). Clinical evaluation further requires that
116 failure modes be weighted by possible harm, as emphasized in medical-answer evaluation frame-
117 works such as Med-PaLM [Singhal et al. \[2023\]](#). MedVIGIL connects these threads by making
118 abstention conditional on visual-channel integrity: the desired behavior is not generic refusal, but

119 doctor-adjudicated safe action for each probe, such as correcting a false premise or stating that an
 120 ROI-masked image no longer supports the answer. Appendix A gives a compact feature matrix
 121 comparing MedVIGIL with these adjacent benchmark families.

122 3 Evidence-Conditional Trustworthiness

123 3.1 Formal Setup

124 Let f denote a medical VLM that maps an image I and a question q to either a textual response
 125 $y = f(I, q)$ or, in the MCQ wrapper, a selected option letter $\hat{\ell} = f_{\text{MCQ}}(I, q, \mathcal{O})$ drawn from a
 126 five-option set $\mathcal{O} = \{A, B, C, D, E\}$. Let $\mathcal{P}$ be a set of evidence-perturbing operators acting on
 127 either the textual or visual channel. A perturbation $\pi \in \mathcal{P}$ may rewrite the question (paraphrase,
 128 negation, specificity-drop, knowledge-only rewrite, or hallucination trap) or replace the image with
 129 a region-of-interest variant (ROI-masked, ROI-only, lr_flip). Let $\rho(y) \in \{0, 1\}$ indicate whether the
 130 open-ended response explicitly refuses, asks for a valid image, or states that the input is inadequate;
 131 in the MCQ wrapper, this behavior is normalized as option E, whose wording is probe-specific and
 132 can express a false-premise correction, insufficient visual evidence, or a conventional refusal.

133 Within this trustworthiness frame, silent failure is the event in which a model answers through broken
 134 evidence rather than marking the evidence as inadequate. For a perturbed probe π whose constructed
 135 correct option is the refusal option $\ell_\pi^* = E$ (e.g., a hallucination trap whose only correct answer is to
 136 reject the false premise), we say that f exhibits silent failure on (I, q, π) when

$$\hat{\ell}_\pi(f) \neq \ell_\pi^* \quad (\text{MCQ form}) \quad \text{or} \quad \rho(f(\pi(I, q))) = 0 \quad (\text{open-ended form}). \quad (1)$$

137 The definition intentionally separates the correctness of a medical answer from the safer prior ques-
 138 tion of whether any answer should be given when the evidence channel has been perturbed.

139 3.2 Probe Families and Output Taxonomies

140 The MedVIGIL evaluation comprises two probe families that together test whether a model recog-
 141 nizes when the textual or visual evidence channel is unreliable. Text-side probes operate on an intact
 142 image but perturb the question: *hallucination traps* embed a false premise that the image contradicts;
 143 *negation* inverts the original question and gold answer; *specificity-drop* removes a clinical qualifier
 144 from the question while keeping the gold answer fixed; *paraphrase (T-CF)* rewords the question
 145 without changing the gold; and *knowledge-only* probes ask a clinically related question that is an-
 146 swerable from text alone. Image-side probes operate on the original question but perturb the visual
 147 channel using a region-of-interest (ROI) bounding box: *ROI-masked* replaces the relevant region
 148 with a uniform mid-grey patch, *ROI-only* masks everything outside the ROI, and *lr_flip* mirrors the
 149 full image left-right, with the gold answer flipped only when the case is laterality-dependent. The
 150 full construction-and-scoring pipeline that turns these probe families into the released benchmark
 151 and into per-model audit results is shown in Figure 2.

152 The output taxonomy treats non-refusal under perturbed evidence as non-binary. We classify API-
 153 level outputs into clean refusal, pedagogical hedge, educational deflection, silent failure, and low-
 154 severity non-refusal. Only clean refusal is unambiguously safe. A pedagogical hedge can acknowl-
 155 edge that the visual evidence is incomplete while listing severe findings in conditional form; an
 156 educational deflection may not acknowledge the perturbation but gives general clinical guidance;
 157 silent failure makes a concrete patient-attributed claim despite the broken evidence contract. In the
 158 MCQ wrapper, option E is not a generic “refuse all” label: it is a doctor-reviewed bucket for the
 159 correct safe action on that probe, such as correcting a false premise or stating that the masked ROI
 160 makes the image insufficient. This exact-letter format intentionally compresses richer open-ended
 161 behavior so that the main leaderboard is auditable without judge-model dependence; the released
 162 open-ended variant preserves the finer taxonomy for qualitative analysis. Appendix F shows a con-
 163 crete case-level table with the image, paired probes, doctor-adjudicated correct options, and model
 164 behavior.

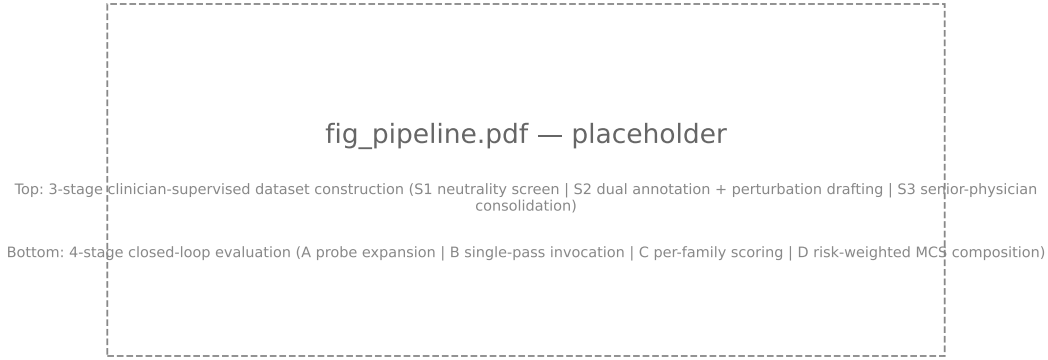


Figure 2: End-to-end MedVIGIL pipeline. **Top:** three-stage clinician-supervised dataset construction (Section 4.4). Two radiologists screen source questions for neutrality, jointly annotate CRT/gold/options/ROI/laterality, and draft every perturbation under clinical-realism constraints; a senior physician consolidates and finalises the released MCQ correct letter. **Bottom:** four-stage closed-loop evaluation (Section 5.1)—deterministic probe expansion, single-pass single-letter MCQ invocation, per-family scoring stratified by CRT/source/modality, and risk-weighted harmonic-mean MCS composition (Eq. 4). The data flow is closed because every released probe answer is the consolidating physician’s; the diagnostic flow is closed via the blind-input controls in Appendix G.

165 4 MedVIGIL: A Dataset-Backed Evaluation Suite

166 4.1 Evaluative Claims and Assumptions

167 MedVIGIL is a dataset-backed evaluation suite for deployment-oriented trustworthiness rather than
 168 a conventional diagnostic leaderboard. Its primary evaluative claim is about *evidence-conditional*
 169 *safe failure*: for a question whose answer requires visual evidence, a medical VLM should select an
 170 explicit refusal option (or, in the open-ended form, refuse, ask for a valid image, or flag the false
 171 premise) when the textual or visual evidence has been perturbed in ways the benchmark constructs.
 172 The suite supports three claims that ordinary medical VQA benchmarks do not isolate. First, a model
 173 can have non-trivial clean-input ability while still failing to recognize a false-premise hallucination
 174 trap. Second, scoring on ROI-only versus ROI-masked variants can distinguish models that use the
 175 answer-relevant image region from models that answer the same way regardless of which image con-
 176 tent is visible. Third, paired counterfactual triplets can separate paraphrase-stable, image-grounded
 177 behavior from paraphrase-brittle or language-prior-driven behavior.

178 The assumptions are explicit. MedVIGIL applies when the evaluated question is vision-required,
 179 the perturbation either rewrites the question with an image-contradicted premise or removes the
 180 answer-relevant region of the image, and safe behavior is to select the constructed refusal option (in
 181 the MCQ form) or an explicit uncertainty/refusal response (in the open-ended form). This scope
 182 distinguishes textual-premise and ROI-conditioned visual perturbations from full-image acquisition
 183 artifacts that may still leave diagnostically useful evidence, and it positions MedVIGIL as a comple-
 184 ment to diagnostic-accuracy, workflow-prevalence, and real-world deployment studies.

185 4.2 Paired Evaluation Design

186 Each case $(I_i, q_i, y_i^*, r_i, t_i)$ contains the image, question, gold answer, clinical risk tier $r_i \in$
 187 $\{L1, \dots, L5\}$, and a binary text-only-answerability flag t_i . The paired probe set evaluates the same
 188 case under each text-side and image-side perturbation π , holding the gold case constant; this isolates
 189 evidence integrity from ordinary question difficulty.

190 The suite has three evidence layers. *Original* probes (clean image, original question) act as a capa-
 191 bility control so that blanket refusal cannot masquerade as robustness. *Perturbation* probes pair each
 192 case with hallucination traps, paraphrase / negation / specificity-drop / knowledge-only rewrites, and
 193 three ROI-conditioned image variants. *Counterfactual triplets* link anchor, T-CF, and V-CF probes

194 into a case-level coherence check. The per-family metrics that score these layers are defined in
195 Section 5.

196 4.3 Source Mix and Risk Coverage

197 MedVIGIL contains 300 cases drawn from four publicly redistributable sources—VQA-RAD
198 (yes/no radiology VQA), SLAKE (bilingual semantic VQA), ROCO (caption-derived prompts), and
199 CXR collections (“most clinically significant finding”)—with one question per case at the mani-
200 fest layer and up to ten evaluation probes per case. Sampling was stratified-quota balanced across
201 L1–L5 risk tiers with a thirty-case minimum per tier; the realised distribution favours L3 (presence-
202 of-finding) and L1 (modality / meta) cases that the source corpora supply abundantly. Per-source
203 counts and the realised CRT distribution are in Appendix B (Table 3).

204 4.4 Three-Stage Annotation Pipeline

205 MedVIGIL is constructed by three clinicians: two attending-level radiologists who annotate in par-
206 allel, and one senior physician who consolidates. The top half of Figure 2 traces the workflow.

207 **Stage 1 — Neutral-question screening.** The two radiologists admit a source question to the mani-
208 fest only if it satisfies two criteria: (i) *image dependence*—the wording cannot be answered from text
209 alone, and (ii) *non-leading phrasing*—the wording does not telegraph the gold via answer-revealing
210 qualifiers or leaky templates. Both criteria are applied jointly during screening and re-verified at
211 Stage 3. This neutrality contract is what lets a later non-refusal under perturbation be attributed to a
212 true evidence-conditional failure rather than to question-template leakage.

213 **Stage 2 — Dual annotation and supervised perturbation drafting.** The two radiologists then
214 independently annotate each surviving case along five dimensions: CRT (L1–L5), gold answer, five-
215 option candidate set including the explicit refusal option, answer rationale plus differential set, and
216 ROI bounding box with laterality flag; per-case inter-annotator agreement is logged. They then
217 jointly draft every released perturbation—paraphrase (T-CF), negation, specificity-drop, knowledge-
218 only rewrite, two hallucination traps, ROI-masked / ROI-only / lr_flip—under clinical-realism con-
219 straints: trap premises invent only conditions a clinician might mis-state in a referral note, ROI
220 masks correspond to plausible image-quality failures rather than synthetic artefacts, and lr_flip re-
221 spects the laterality flag. No perturbation is released without radiologist sign-off.

222 **Stage 3 — Senior-physician consolidation.** A third senior physician adjudicates disagreement, fi-
223 nalises the gold, CRT, text-only-answerability flag, ROI box, and the MCQ correct letter, and rejects
224 any drafted perturbation that violates the construction invariants—paraphrase preserves the gold,
225 negation inverts it, knowledge-only probes remain image-independent, hallucination traps contain a
226 premise the image contradicts, ROI-masked probes remove the answer-relevant region, and triplet
227 anchors satisfy $f(\text{anchor}) = f(\text{T-CF}) \neq f(\text{V-CF})$. Probes that fail any invariant are dropped, not
228 patched. The consolidated case is the released artefact—every safety tier, gold, perturbation, and
229 refusal option is traceable to a named clinician role.

230 The full per-case schema and the contents of the risk, grounding, and triplet annotation layers are
231 documented in Appendix B (Table 4).

232 Each case is also wrapped in an anchor / T-CF / V-CF counterfactual triplet whose joint invariant
233 $f(\text{anchor}) = f(\text{T-CF}) \neq f(\text{V-CF})$ scores grounding via paraphrase brittleness and counterfactual
234 blindness; the triplet construction and the release/redistribution boundary that ships the case JSONs,
235 scoring code, and cached outputs independently of credentialed images are detailed in Appendix B.

236 5 Metrics

237 The probe families (Section 3) are scored with eight headline metrics in the multiple-choice form—
238 the model selects a letter from the doctor-defined five-option wrapper, scoring is exact, and no judge
239 model is used. A paired open-ended variant is released as an auxiliary artefact for qualitative analysis
240 only.

241 5.1 Closed-Loop Evaluation Pipeline

242 Four deterministic stages close the loop from the case manifest to the audit composite (bottom half of
 243 Figure 2): **(A)** probe expansion deterministically replays the perturbations of Section 4.4 into a five-
 244 option MCQ container; **(B)** each VLM is invoked once per probe with a single-letter A–E prompt,
 245 with no judge model, chain-of-thought, or multi-sample voting; **(C)** the selected letter is matched
 246 against the doctor-finalised correct letter and reduced to the per-family metrics below, stratified
 247 by CRT, source, modality, and text-only-answerability; **(D)** harm-scaled weights aggregate per-tier
 248 SFR into SFR_w (Eq. 2), which becomes the Safety axis of the harmonic-mean MCS (Eq. 4). The
 249 pipeline is closed in two senses—the data flow because every released answer is the consolidating
 250 physician’s, and the diagnostic flow because the blind-input controls in Appendix G let any score
 251 gap or non-gap be attributed to language-prior reliance or grounding failure. Stage-by-stage detail
 252 is in Appendix C.

253 5.2 Per-Family Metric Definitions

254 Let $\hat{\ell}_j(f) \in \{A, \dots, E\}$ denote model f ’s selected letter on probe j , and let ℓ_j^* denote the doctor-
 255 finalised correct letter. All metrics below are exact-match aggregates of the binary correctness signal
 256 $\mathbb{I}[\hat{\ell}_j(f) = \ell_j^*]$, computed from a single per-probe letter trace.

257 *Capability and text robustness.* *Overall accuracy* averages the correctness signal across all probes;
 258 *Original accuracy* $\text{Acc}_{\text{orig}}(f)$ restricts the average to clean (image, original-question) probes. Three
 259 text-robustness scores isolate stability under wording-only perturbations whose gold either remains
 260 unchanged or transforms in a controlled way: *paraphrase robustness* $\text{PR}(f) = \Pr_j[\hat{\ell}_j^{\text{orig}} = \hat{\ell}_j^{\text{tcf}}]$
 261 measures anchor–T-CF letter agreement; *negation consistency* $\text{NEG}(f)$ is accuracy on negated ques-
 262 tions whose gold flips with the inversion; *specificity-drop robustness* $\text{SDR}(f)$ is accuracy after one
 263 clinical qualifier is removed from the question, with the gold unchanged.

264 *Evidence safety.* The hallucination-trap probe wraps a false-premise question whose only correct
 265 option is the explicit refusal letter “the premise of this question is incorrect.” The *silent failure rate*
 266 $\text{SFR}(f) = \Pr_{j \in \text{trap}}[\hat{\ell}_j(f) \neq \ell_j^*]$ is the fraction of trap responses that pick a non-refusal option;
 267 lower values are safer. A clinical-risk-weighted variant SFR_w feeding the composite is defined in
 268 Section 5.3.

269 *Visual grounding.* For ROI-annotated cases, the same question is scored on two image variants:
 270 *ROI-only* retains only the answer-relevant region (the rest replaced by mid-grey), and *ROI-masked*
 271 removes only the answer-relevant region. The case-paired contrast $\text{VGR}(f) = \text{Acc}_{\text{ROI-only}}(f) -$
 272 $\text{Acc}_{\text{ROI-masked}}(f)$ is signed: positive values indicate that accuracy is higher when the answer-relevant
 273 region is visible, negative values can indicate prior-driven answering or instability under masking.
 274 VGR therefore must be read together with ROI-masked accuracy, whose doctor-adjudicated correct
 275 option is the explicit refusal letter.

276 *Language-prior and counterfactual coherence.* *Language-prior leakage* $\text{LPL}(f)$ is accuracy on
 277 knowledge-only probes whose answer is recoverable from clinical knowledge alone; LPL is a capa-
 278 bility control and only carries a safety implication when paired with a non-degenerate VGR. *Triplet*
 279 *coherence* $\text{TR-coh}(f)$ is the fraction of counterfactual triplets that simultaneously satisfy paraphrase
 280 preservation and counterfactual sensitivity, $\hat{\ell}^{\text{anchor}} = \hat{\ell}^{\text{T-CF}} \neq \hat{\ell}^{\text{V-CF}}$.

281 5.3 The MedVIGIL Composite Score

282 The composite combines three named trustworthiness axes—Capability, Safety, and Grounding—
 283 into a single bottleneck-sensitive summary, visualised in Figure 3. A naive aggregate SFR would
 284 weight an L1 modality trap and an L5 “don’t-miss” trap equally, which is the wrong rule for a
 285 deployment-oriented medical audit. We therefore define a clinical-risk-weighted silent failure rate
 286 as the harm-weighted average of the per-tier rates.

$$\text{SFR}_w(f) = \frac{\sum_{k=1}^5 w_k \text{SFR}_{L_k}(f)}{\sum_{k=1}^5 w_k}, \quad (w_1, \dots, w_5) = (1, 2, 3, 5, 8). \quad (2)$$

287 The weights place an L5 silent failure at $8\times$ the weight of an L1 silent failure, tracking the harm-if-
 288 wrong scale that defines the tiers. SFR_w is reported as a separate column in Table 6 and acts as the
 289 Safety input to the composite.

290 The three component axes are then defined on the $[0, 100]$ scale.

$$\begin{aligned}\text{Cap}(f) &= \frac{1}{4}(\text{Acc}_{\text{orig}}(f) + \text{PR}(f) + \text{NEG}(f) + \text{SDR}(f)), \\ \text{Safe}(f) &= 100 - \text{SFR}_w(f), \\ \text{Ground}(f) &= \frac{1}{2}(\text{clip}(\text{VGR}(f) + 50, 0, 100) + \text{Acc}_{\text{ROI-masked}}(f)).\end{aligned}\quad (3)$$

291 The Capability axis averages original-probe accuracy with the three text-robustness scores. The
 292 Safety axis is the complement of SFR_w . The Grounding axis pairs the signed ROI contrast (shifted
 293 and clipped onto $[0, 100]$) with ROI-masked accuracy; the latter has the explicit refusal letter as its
 294 doctor-adjudicated correct option, so VGR alone is never treated as a safety score.

295 The MedVIGIL Composite Score is the harmonic mean of these three axes.

$$\text{MCS}(f) = \frac{3 \text{Cap}(f) \text{Safe}(f) \text{Ground}(f)}{\text{Cap}(f) \text{Safe}(f) + \text{Cap}(f) \text{Ground}(f) + \text{Safe}(f) \text{Ground}(f)}.\quad (4)$$

296 The harmonic-mean form penalises the weakest component—a 90/90/10 model receives $\text{MCS} \approx 25$
 297 rather than 63—so high clean-input capability cannot offset weak grounding or unsafe trap be-
 298 haviour. Required reporting axes (probe kind, modality, CRT, source, text-only-answerability) and
 299 deployment-licence caveats are deferred to Appendix C.

300 6 Empirical Audit

301 6.1 Setup

302 The audit is aggregated from per-model baseline JSONL outputs scored against the MCQ wrapper
 303 of every probe in MedVIGIL. It covers 16 vision-capable model configurations across five propri-
 304 etary providers and two open-weight medical-specific systems (OpenAI: GPT-5.5, GPT-5.4, GPT-
 305 4o, GPT-5.4-mini, GPT-5.4-nano; Anthropic: Claude Opus-4.7, Claude Sonnet-4.6, Claude Haiku-
 306 4.5; Google: Gemini 3-Flash, Gemini 3.1-Flash-Lite; Alibaba/Qwen via Together: Qwen3.5-9B,
 307 Qwen3.5-397B-A17B; Moonshot AI via Together: Kimi-K2.5, Kimi-K2.6; locally hosted LLaVA-
 308 Med-v1.5-mistral-7B and HuatuoGPT-Vision-7B), with up to 2,556 scored probes per model. The
 309 evaluation includes original image-question probes, text-counterfactual paraphrases, negated ques-
 310 tions, specificity-drop prompts, knowledge-only controls, hallucination traps, left-right flips, and
 311 ROI-only versus ROI-masked visual controls.

312 Blind-input controls are consolidated in Appendix G: two external DeepSeek text-only baselines and
 313 matched no-image ablations for selected GPT, Claude, and Gemini systems. We keep these controls
 314 separate from the deployed vision-capable runs because they diagnose language-prior reliance rather
 315 than image-present performance.

316 6.2 Headline Results

317 Table 1 summarises the main result matrix; grouped headers separate capability, evidence safety,
 318 text robustness, language prior, and the MCS composite.

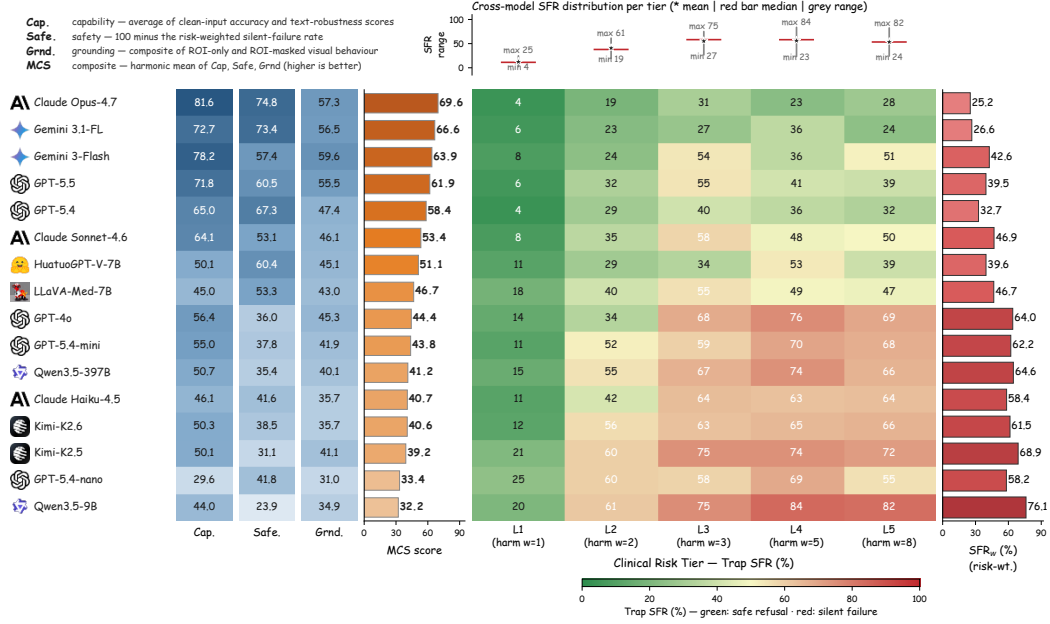


Figure 3: MedVIGIL audit summary. **Left:** MCS component decomposition—Capability, Safety ($100 - \text{SFR}_w$), and Grounding cells plus harmonic-mean MCS score, models ordered by MCS. **Right:** Risk-tier silent-failure heatmap. Centre matrix is per-tier trap SFR (L_1 – L_5); left blue bar is original-probe accuracy; top strip shows cross-model min/median/max SFR per tier; right red bar is SFR_w (Eq. 2). Models sorted by SFR_w , safest at top.

Table 1: Headline MedVIGIL results (%; VGR in pp). ↓ lower is safer, ↑ higher is better. †HuatuoGPT-V was trained on PubMedVision (overlaps VQA-RAD/SLAKE).

Model	Capability		Evidence Safety		Text Robustness			Prior	Composite
	Overall	Original	SFR↓	VGR	PR	NEG	SDR		
5.5	69.4	75.0	36.8	+12.0	83.7	68.9	59.8	99.6	61.9
5.4	65.2	68.0	28.8	+15.6	75.7	56.4	59.8	98.6	58.4
4o	56.1	59.3	53.0	+3.6	70.0	45.8	50.6	98.6	44.4
5.4-mini	54.7	55.3	49.7	-8.9	72.7	44.9	47.0	98.2	43.8
5.4-nano	38.8	31.7	51.7	-27.1	45.0	16.9	25.0	90.5	33.4
AI Opus-4.7	76.5	78.7	22.0	+22.9	84.7	83.1	79.9	98.6	69.6
AI Sonnet-4.6	61.2	63.3	41.3	+18.2	75.3	60.9	56.7	97.9	53.4
AI Haiku-4.5	49.6	46.7	49.0	-9.4	63.3	32.4	42.1	97.5	40.7
3-Flash	71.9	77.0	37.2	+41.1	84.0	75.6	76.2	98.9	63.9
3.1-Flash-Lite	70.7	73.0	22.3	+49.5	84.0	63.1	70.7	98.9	66.6
Qwen3.5-9B	39.4	43.3	62.8	+12.5	72.0	28.4	32.3	84.5	32.2
Qwen3.5-397B-A17B	46.8	53.0	54.3	+24.0	73.7	33.3	42.7	92.2	41.2
Kimi-K2.5	47.3	49.7	60.0	+10.4	72.7	36.4	41.5	95.1	39.2
Kimi-K2.6	46.6	49.3	50.8	-2.1	78.0	36.0	37.8	95.4	40.6
LLaVA-Med-7B	44.8	43.3	43.0	+28.6	70.3	15.6	50.6	75.6	46.7
HuatuoGPT-V-7B†	55.5	55.3	31.5	+28.1	67.7	26.2	51.2	91.9	51.1

Three observations follow from the table and component visualization. First, the individual metrics do not collapse to a single ranking across all 16 vision-capable models: Claude Opus 4.7 leads on accuracy (Overall 76.5%), Gemini 3.1 Flash-Lite has the largest ROI contrast (VGR +49.5 pp), GPT-5.5 leads knowledge-only accuracy (LPL 99.6%), and aggregate SFR is lowest for Claude Opus 4.7

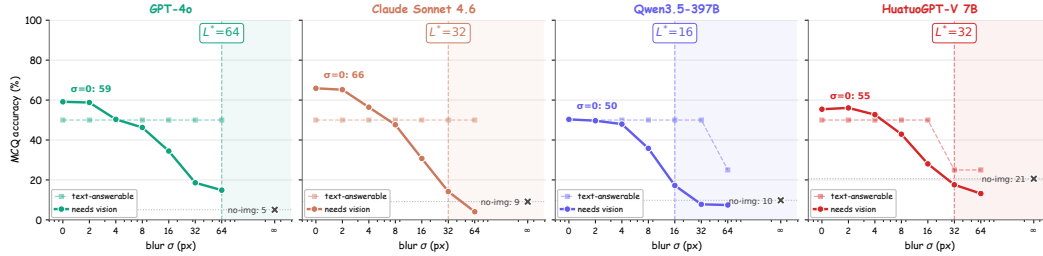


Figure 4: Visual information decay on the 300 image-required cases. Solid curves track MCQ accuracy as the blur σ grows from 0 to 64 px and finally to no-image ($\sigma=\infty$, X marker); dashed curves are the text-answerable control subset whose answers do not require the image. The control stays flat across all σ , attributing the solid-curve collapse to lost visual evidence rather than degraded model competence. The shaded region right of L^* is the language-prior-dominated regime in which $\geq 80\%$ of visual contribution is gone. Per-risk-tier curves and the full numerical table are deferred to Appendix H.

(22.0%) and Gemini 3.1 Flash-Lite (22.3%) at different accuracy levels. Second, accuracy does not imply trustworthy behavior under broken evidence: Gemini 3 Flash reaches 71.9% Overall but 37.2% SFR, GPT-4o reaches 56.1% Overall but 53.0% SFR, and Kimi-K2.5 reaches 47.3% Overall but 60.0% SFR; smaller and emerging multimodal systems cluster at the lower end of MCS when risk-weighted trap silent-failure or ROI grounding is weak (Appendix E, Table 6). Third, Figure 3 shows why MCS changes the ranking: Claude Opus 4.7 leads because it is strong on all three component scores, Gemini 3.1 Flash-Lite ranks second because its Safety and Grounding scores are both high, GPT-4o falls to 44.4 because Safety is its bottleneck, and Qwen3.5-9B remains lowest because its Safety component is below 25.

6.3 Detailed Audit Breakdowns

Three further breakdowns are deferred to Appendix E: per probe kind (Table 5, including the visual-control diagnostic where some models score *higher* when the ROI is masked than when it is shown); per clinical risk tier (Table 6, the table that defines the per-tier SFR values aggregated into SFR_w); and per source dataset (Table 7, used to confirm that the headline ordering is not driven by a single corpus). Appendix D interprets these patterns in more depth.

6.4 Ablation: Visual Information Decay

The blind-input controls in Appendix G answer the binary question *can the model answer with the image suppressed?* They do not say *at what point* of visual degradation a model stops using the image and starts answering from language priors. To localise that crossover we sweep an isotropic Gaussian blur over each of the 300 case images at $\sigma \in \{0, 2, 4, 8, 16, 32, 64\}$ px and compare to the no-image control ($\sigma=\infty$), evaluating four representative models (🌀 GPT-4o, 🐼 Claude Sonnet 4.6, 🦾 Qwen3.5-397B-A17B, 🤖 HuatuoGPT-V 7B) on the 300 *original* MCQ probes. We define the language-takeover point L^* as the smallest blur at which the residual visual contribution drops below 20% of the clean-image contribution, $(\text{acc}_\sigma - \text{acc}_\infty)/(\text{acc}_0 - \text{acc}_\infty) \leq 0.2$.

Three patterns appear (Figure 4 and Table 11 in Appendix H). First, the text-answerable control is flat across σ for every model, attributing the solid-curve collapse to lost visual evidence rather than to a generic competence drop. Second, L^* separates the four models by a factor of four ($16 \rightarrow 64$ px): GPT-4o is the most visually anchored ($L^*=64$); Claude Sonnet 4.6 has the highest clean accuracy and largest visual contribution ($65.9\% \rightarrow 9.1\%$, drop 56.8 pp) but falls back to language priors at $\sigma=32$; Qwen3.5-397B has the smallest visual contribution (40.5 pp) and the earliest takeover ($L^*=16$); and HuatuoGPT-V 7B has the highest no-image floor (20.6%, four times that of GPT-4o), suggesting that domain-specific medical fine-tuning concentrates clinical knowledge in the language backbone rather than the visual aligner. Third, the curves do not follow the headline MCS ordering—a model that uses vision robustly under degradation can still rank below a model that uses vision well only on intact inputs—which is consistent with the claim that visual grounding is a separate trustworthiness axis.

7 Limitations

MedVIGIL focuses on one component of deployment-oriented trustworthiness: unsafe non-refusal under controlled perturbations of the textual or visual evidence channel. Its scope is complementary to clinical diagnostic accuracy, deployment readiness, and hospital-workflow prevalence studies. Several specific scoping limits apply. First, the visual perturbations evaluated here are ROI-conditioned (ROI-masked, ROI-only) and a global laterality flip; full-image corruption operators such as missing, blank, noise, blur, and mismatched-image inputs are not part of the current empirical audit. Second, the visual counterfactual is implemented as a textual conditional anchor rather than a regenerated image, so the V-CF measures hypothetical-conditional reasoning rather than visual response to a fully synthesized counterfactual scene. Third, the severity-aware metrics (Critical Findings Hallucination Rate, Severity Drift Magnitude) and the patch-masking Visual Faithfulness Rate are part of the broader MedVIGIL design but are not reported here because they require severity scoring and counterfactual image generation that are not in this audit. Fourth, the ROI bounding boxes are adjudicated benchmark regions used to position binary masks, not pixel-level segmentation labels or a localization benchmark. Fifth, although two radiologists annotate every case and a third physician adjudicates the final manifest, the current audit does not include an independent human-baseline experiment in which clinicians answer the same MCQ traps as the models. Sixth, the source images come from public or publicly redistributable VQA-style corpora, so overlap with model pretraining cannot be ruled out and the audit should not be read as evidence on private hospital cohorts. Seventh, we do not yet ablate thinking mode, temperature, safety-instruction prompts, or medical-specialized model families; those studies are natural uses of the released probe schema. The refusal classifier and language-prior leakage proxy are transparent, auditable benchmark instruments rather than clinical measurement instruments.

8 Conclusion

Trustworthy medical VLM evaluation cannot stop at intact-image accuracy. A model may answer clean images correctly while still accepting false premises, ignoring ROI-conditioned evidence loss, or relying on language priors when the image should be decisive. MedVIGIL operationalizes deployment-oriented trustworthiness through paired clean, textual-perturbation, ROI-conditioned, and clinical-risk-stratified probes, with silent failure serving as the key measured failure mode under broken evidence. The audit shows that accuracy, safe refusal, and visual grounding do not reduce to a single leaderboard, motivating the MedVIGIL Composite Score as a compact summary alongside the underlying per-axis metrics. Medical VLM evaluations should therefore test not only whether a model can answer an intact image, but also whether it can recognize when the evidence contract has failed.

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A Benchmark Positioning

Table 2 summarizes how MedVIGIL differs from adjacent benchmark families. The key distinction is not only the medical domain, but the combination of paired evidence perturbations, doctor-adjudicated safe targets, and risk-weighted silent-failure scoring.

Table 2: Positioning of MedVIGIL against adjacent benchmark families. ✓ marks a primary evaluation target; ◦ marks partial or indirect coverage; – marks a non-central target.

Benchmark family	Intact medical QA	Valid-input hallucination	Nuisance shift	Generic abstention	Broken evidence contract	Risk-weighted silent failure
Medical VQA/VLM	✓	–	–	–	–	–
Hallucination	–	✓	–	–	–	◦
Robustness/corruption	◦	–	✓	–	–	–
Refusal/abstention	–	–	–	✓	–	◦
MedVIGIL (ours)	✓	◦	◦	✓	✓	✓

Representative prior families include medical VQA/VLM benchmarks [Lau et al. \[2018\]](#), [Liu et al. \[2021\]](#), [Zhang et al. \[2023\]](#), [Li et al. \[2023a\]](#), hallucination benchmarks [Rohrbach et al. \[2018\]](#), [Li et al. \[2023b\]](#), [Chen et al. \[2024\]](#), robustness benchmarks [Hendrycks and Dietterich \[2019\]](#), [Shah et al. \[2019\]](#), [Gokhale et al. \[2020\]](#), and abstention work [Geifman and El-Yaniv \[2017\]](#), [Ren et al. \[2023\]](#). MedVIGIL includes clean-input controls and false-premise traps, but its central target is the broken medical evidence contract rather than free-form hallucination under an otherwise valid visual input.

B Dataset Construction Details

This appendix expands the dataset-side details that Section 4 compresses into forward pointers: the source mix and CRT coverage table, the per-case object schema, the three annotation layers (risk, grounding, triplet) attached during Stage 2 of the construction pipeline (Section 4.4), the counterfactual-triplet construction, and the artefact-boundary policy that governs release.

Table 3: Source mix and realised CRT coverage of MedVIGIL. CRT counts are doctor-adjudicated.

Source	Cases	Primary role
VQA-RAD	120	Yes/no and short-answer radiology VQA; cross-comparable baseline
SLAKE	60	Semantically labeled, knowledge-grounded medical VQA
ROCO	60	Radiology caption-derived open-ended questions
CXR collections	60	“Describe the most clinically significant finding” on chest X-rays
Total	300	Realised CRT distribution L1:L2:L3:L4:L5 = 71:31:118:43:37

B.1 Per-Case Schema

The case schema makes the evaluation contribution explicit. Each released case stores: (i) source provenance from one of the four source datasets in Table 3, with original identifier and a relative image path; (ii) the question, gold answer, candidate answer options, and three semantically equivalent answer variants used for relaxed open-ended scoring; (iii) the radiologist-annotated and physician-adjudicated clinical risk tier and text-only-answerability flag; (iv) a grounding annotation with answer rationale, ROI bounding box, differential set, and laterality-dependence flag; (v) the probe battery (paraphrase, negation, specificity-drop, knowledge-only, two hallucination traps with explicit false-premise descriptions, and three image-side perturbations); and (vi) the counterfactual triplet anchor used to compute triplet coherence. Table 4 lists the released components.

Table 4: Core case object in MedVIGIL. The fields are designed so that the same case supports MCQ scoring, open-ended scoring, language-prior analysis, counterfactual grounding, and release provenance.

Component	Key fields and purpose
Source	Source dataset (VQA-RAD, SLAKE, ROCO, or CXR collection), original case identifier, and image relative path.
Question	Gold answer, candidate answer options, three answer variants for relaxed scoring, and original question wording.
Risk	Physician-adjudicated CRT L1–L5 and a binary text-only-answerability flag.
Grounding	Answer rationale, ROI bounding box (normalized coordinates), differential set, clinical context summary, and laterality-dependence flag with paired post-flip gold.
Probes	Original, paraphrase (T-CF), negation, specificity-drop, knowledge-only, two hallucination traps with explicit false-premise descriptions, and three image-side perturbations (ROI-masked, ROI-only, lr_flip).
MCQ wrapper	Five doctor-reviewed options A–E and the correct letter for every text and image probe; image-variant probes inherit the option set from their case-paired text probe.
Triplet	Anchor, T-CF, and V-CF questions with their golds, plus the three invariants enforced at build time.

496 B.2 Risk and Grounding Annotation Layers

497 Each released case carries three annotation layers, all attached during Stage 2 of the pipeline (Sec-
498 tion 4.4) and finalised at Stage 3.

499 The *risk layer* stores the clinical risk tier (CRT) together with the binary text-only-answerability flag.
500 The CRT is one of L1 (meta or modality questions, no clinical harm if wrong), L2 (anatomical locali-
501 sation with no management impact), L3 (presence of finding, delayed care if missed), L4 (significant
502 pathology or characterisation, wrong treatment if missed), or L5 (time-critical, “don’t-miss” find-
503 ings). It exists so that no failure mode is treated as harm-equivalent: a wrong modality-recognition
504 answer is not equated with a missed intracranial haemorrhage or a fabricated myocardial infarction,
505 and the same risk weighting feeds the audit composite (Section 5.3). The text-only-answerability
506 flag records whether a clinically informed reader could plausibly answer the question from word-
507 ing alone. Stage 1 already removes questions whose wording trivially leaks the answer; this flag is
508 the finer-grained marker that residually-ambiguous cases carry, and it lets analyses condition on a
509 continuous visual-grounding requirement.

510 The *grounding layer* stores the answer rationale, an ROI bounding box (used to construct the ROI-
511 masked and ROI-only image perturbations), a differential set of plausible alternatives, a clinical
512 context summary, and a laterality-dependence flag paired with a post-flip gold answer. The ground-
513 ing layer is what links the text-side probes (which read from the gold and differential set) to the
514 image-side probes (which read from the ROI box and laterality flag); without it, ROI masking and
515 laterality flipping would have to be guessed from raw images. It is also what allows a single audit to
516 stratify trustworthiness failures along both clinical harm *and* visual necessity, rather than reporting
517 a single aggregate error rate.

518 B.3 Counterfactual Triplets

519 Counterfactuals are the mechanism that turns MedVIGIL from a static VQA set into a grounding
520 evaluation. Each case is wrapped in a triplet that consists of an anchor, a textual counterfactual (T-
521 CF), and a textual visual counterfactual (V-CF). The anchor is the original (image, question, gold)
522 probe. The T-CF rewrites the question with semantically equivalent wording while keeping the
523 image and gold answer fixed; a model that flips between the anchor and the T-CF is not paraphrase-
524 robust. The V-CF leaves the image fixed but augments the question with a single counterfactual
525 condition (for example, “Hypothetically, if no consolidation were present:”) and replaces the gold

with the answer that becomes correct under that condition. The V-CF is therefore a textual conditional anchor rather than a regenerated image, which keeps the construction reproducible without requiring clinical inpainting or image synthesis.

Three invariants are enforced at build time: the T-CF question must differ from the anchor question, the T-CF gold must match the anchor gold, and the V-CF gold must differ from the anchor gold. Cases that fail any invariant are dropped, not patched. Of the 300 cases in the manifest, 240 yield triplets that satisfy all three invariants; the remaining 60 are predominantly L1 modality or meta questions for which a clean visual counterfactual cannot be constructed without fabricating image content. A grounded model should produce $f(\text{anchor}) = f(\text{T-CF}) \neq f(\text{V-CF})$; deviations decompose into paraphrase brittleness ($f(\text{anchor}) \neq f(\text{T-CF})$) and counterfactual blindness ($f(\text{anchor}) = f(\text{V-CF})$).

B.4 Artefact Boundary and Responsible Release

The artefact boundary separates the evaluation contribution from image redistribution. Sources that permit redistribution can be packaged directly with attribution. Credentialed or competition-governed sources are represented by source pointers and reconstruction scripts. The case JSON, questions, CRT/VGR labels, grounding fields, counterfactual definitions, prompts, scoring code, and cached outputs are the primary evaluation artefacts and can be distributed independently of restricted images.

The release package is organised around a benchmark card, versioned case manifests, executable evaluation code, cached outputs, sample cases for reviewer inspection, and machine-readable dataset metadata. The metadata records licence information, source provenance, sensitive medical-data handling, known limitations and biases, intended and disallowed uses, the two-radiologist-plus-physician adjudication procedure, inter-rater agreement before adjudication, and a maintenance policy with immutable case identifiers and versioned corrections. For medical sources that require credentialed access, the paper and metadata state the access procedure, why gating is necessary, and which parts of the evaluation remain inspectable without protected images.

C Evaluation Pipeline and MCS Details

This appendix expands the four-stage evaluation pipeline summarised in Section 5.1, and the MCS-interpretation notes that Section 5.3 compresses into a forward pointer.

C.1 Stage-by-Stage Detail

Stage A — Probe expansion. The 300-case manifest is expanded into the probe set by replaying the perturbations finalised in Section 4.4: each case yields one *original* probe plus its case-paired text-side perturbations (paraphrase / T-CF, negation, specificity-drop, knowledge-only, two hallucination traps) and three image-side perturbations (ROI-masked, ROI-only, lr_flip). Each probe is wrapped in a five-option MCQ container whose explicit refusal option is the doctor-defined “insufficient evidence / false-premise” answer. Expansion is deterministic: the same manifest plus the same perturbation set always yields the same probe list and the same release-version digest.

Stage B — Single-pass model invocation. Each evaluated VLM is queried once per probe under a fixed prompt template that requests a single letter A–E. No judge model, chain-of-thought scoring, multi-sample voting, or self-consistency aggregation is applied. This gives every audited model the same number of forward passes per case and rules out evaluator-rigging failure modes (inflated scores from biased judges, prompt-tuned scorers, or sampling that amplifies stochastic confidence). The blind-input controls in Appendix G re-run the same prompts with the image suppressed, so any vision-capable score is directly comparable to its no-image counterpart.

Stage C — Per-family scoring. The selected letter is matched against the doctor-finalised correct letter to produce a binary correctness signal, which is reduced to the per-family metrics defined in Section 5: Acc, SFR, VGR, PR, NEG, SDR, LPL, and triplet coherence. Stratified versions by CRT, source dataset, modality, and text-only-answerability flag are produced from the same letter trace, so every figure in Section 6 and every appendix breakdown table is computed from one per-probe record rather than from independent re-evaluations.

Stage D — Risk-weighted composition. The per-family metrics feed two summarisation layers. SFR_w aggregates trap-SFR across CRT with harm-scaled weights (Eq. 2); SFR_w becomes the Safety axis of MCS (Eq. 4). The composite is bottleneck-sensitive by construction: a model is only trustworthy if it is simultaneously capable, safe under risk-weighted traps, and ROI-grounded. The Capability and Grounding axes are themselves audit-traceable composites of probe-level signals, so any low MCS score can be decomposed back to a specific probe family.

C.2 Required Reporting Axes

A MedVIGIL report stratifies every metric by probe kind, modality, CRT, source dataset, and text-only-answerability. This is critical because a single aggregate SFR can hide the distinction between a harmless L1 modality question and a high-risk L5 emergency diagnosis. The paired triplet design also supports within-case comparisons: clean accuracy, paraphrase robustness, and counterfactual sensitivity can be measured on the same underlying clinical example rather than across unmatched samples.

C.3 What MCS Does and Does Not Certify

MCS is an evaluation-side composite, not a clinical-deployment licence. A high MCS is not a certification for any specific care setting; a low MCS indicates that the model is not yet trustworthy under at least one audited evidence-failure condition. The three component scores keep the composite auditable: if a model has a low MCS, Figure 3 shows whether the bottleneck is Capability, Safety, or Grounding.

D Extended Discussion

D.1 Evidence-Contract Failure Is Not Ordinary Hallucination

The main empirical pattern is that deployment-oriented trustworthiness is not reducible to ordinary medical VQA accuracy. Existing hallucination benchmarks ask whether a model adds unsupported content when the image is otherwise valid; robustness benchmarks ask whether a model preserves the answer under nuisance perturbation; abstention benchmarks ask whether the model declines low-confidence cases. MedVIGIL targets the intersection of these settings: the input can still look like a legitimate medical VQA request, but the visual or textual evidence contract has failed. In that regime, a fluent answer is not merely an incorrect answer; it can hide that the upstream evidence was missing, masked, or contradicted by the prompt.

This distinction explains why clean-input capability and evidence-safety behavior diverge in the audit. Claude Opus 4.7 is both highly accurate and comparatively safe, but other models do not follow a single ordering: Gemini 3 Flash remains highly capable while retaining 37.2% aggregate SFR, and GPT-4o combines moderate overall accuracy with 53.0% SFR. The clinical-risk breakdown sharpens the point: GPT-4o has strong L1 original accuracy but reaches 68.2% SFR on L3 traps, 75.6% on L4 traps, and 68.9% on L5 traps. These failures would be obscured by a clean-accuracy-only benchmark.

D.2 Mechanisms Made Testable by Paired Probes

The benchmark is designed to turn plausible failure mechanisms into measurable strata. First, when visual evidence is uninformative, a model may fall back to prompt-conditioned language priors; knowledge-only probes and the DeepSeek text-only baseline bound this contribution. Second, instruction tuning may reward clinical helpfulness more strongly than refusal under broken evidence; hallucination traps test this directly by offering a doctor-reviewed safe option. Third, prior-driven medical text may drift toward severe or abnormal findings, which is why CRT-stratified SFR is reported rather than a single unweighted trap score. Fourth, a model may not rely on the answer-relevant region even when it appears visually capable; the ROI-only versus ROI-masked contrast operationalizes this as VGR.

These mechanisms are not claimed as proven causal explanations. The current evidence establishes that they are worth isolating because the observed metrics disagree: aggregate accuracy, SFR, VGR,

language-prior leakage, and CRT-stratified failures do not collapse to the same ranking. The value of the paired design is therefore falsifiability. A stronger model should maintain original-probe accuracy, reduce trap SFR, preserve answers under paraphrase, change behavior under valid counterfactuals, score higher on ROI-only than ROI-masked probes for vision-required cases, and select the insufficient-evidence option when the ROI has been removed.

D.3 Reading MCS Without Hiding the Components

The MedVIGIL Composite Score is intended as a compact trustworthiness summary, not a replacement for the underlying metrics. Its harmonic form prevents Capability, Safety, and Grounding from substituting for one another: a model with strong ordinary accuracy but weak evidence-safety behavior should not look trustworthy, and a model with strong refusal behavior but weak clean-input capability should not look useful. This resolves otherwise confusing comparisons in the leaderboard. Claude Opus 4.7 leads MCS because it is balanced across the three components; Gemini 3.1 Flash-Lite ranks strongly because its Safety and Grounding scores offset a lower raw Overall accuracy; GPT-4o drops in MCS because Safety is its bottleneck despite nontrivial clean-input performance.

MCS should therefore be read alongside the named components and the risk-tier table. The component figure identifies which dimension fails, while CRT-stratified SFR shows whether failures concentrate in high-risk cases. This is the reason MedVIGIL reports both a composite and auditable axes: the composite supports model-level comparison, but the underlying metrics preserve the evidence needed to decide what kind of intervention is required.

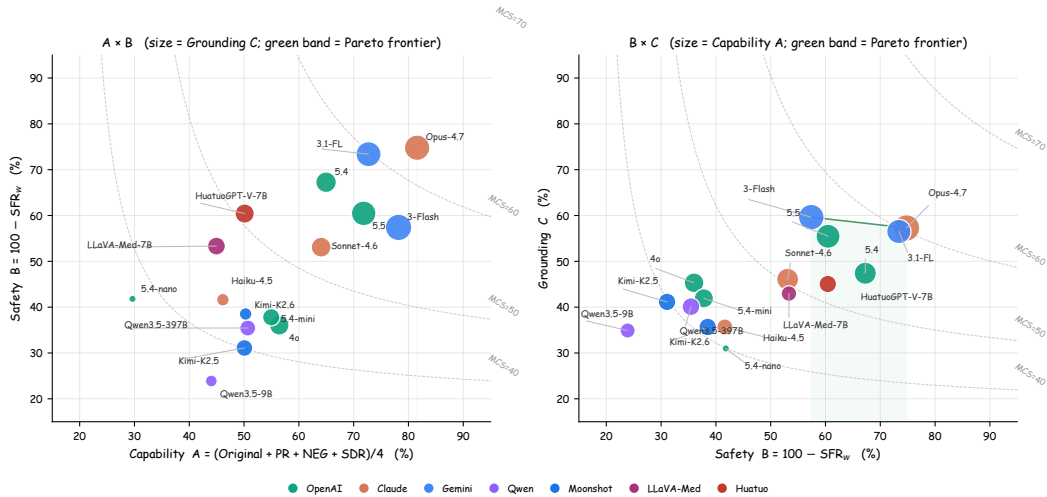


Figure 5: MCS components in two pairwise projections (geometric companion to Figure 3). **Left:** Capability (A) vs Safety ($B = 100 - \text{SFR}_w$); marker size $\propto$ Grounding (C). **Right:** Safety vs Grounding; marker size $\propto$ Capability. Dashed grey curves are MCS isolates at the median value of the unshown component; the green band traces the empirical Pareto frontier on each pair.

D.4 Implications for Evaluation and Deployment

For evaluation, the main implication is that medical VLM audits should include evidence-contract controls rather than treating them as generic robustness checks. A minimal report should include clean-input capability, trap SFR, risk-weighted SFR, ROI-conditioned grounding, language-prior controls, and a paired coherence measure. Reporting only accuracy rewards systems that answer through broken evidence; reporting only refusal rewards systems that avoid useful clinical questions. The paired design constrains both failure modes.

For deployment, the safest mitigation is external input validation before model inference. A clinical pipeline should check image presence, file integrity, resolution, entropy, modality consistency, and expected metadata before forwarding a request to the VLM. If these checks fail, the system should return a structured refusal rather than asking the model to infer whether the image is usable. Model-

side training can still help, but it should include refusal-aware probes that resemble the false-premise and ROI-masked structures used in MedVIGIL, so that the model learns to mark broken evidence rather than answer through it.

E Detailed Audit Breakdowns

This appendix expands the audit results that Section 6 compresses into a single forward pointer: per probe kind (Table 5), per clinical risk tier (Table 6), and per source dataset (Table 7). The headline composite numbers in Section 6 are aggregations of the cells in these three tables.

E.1 Probe-Level Breakdown

Table 5 expands the aggregate metrics by probe kind. This table makes the language-prior issue explicit: knowledge-only accuracy is between 90.5% and 99.6% wherever measured. That result is useful as a capability control, but it also shows why clean VQA accuracy alone cannot certify visual use. A model can answer many medical questions from textual priors while still failing the visual-integrity contract.

Table 5: Accuracy (%) by probe kind. TCF: text-counterfactual paraphrases; NEG: negation; SDR: specificity-drop; LR flip: left-right flip. Missing cells: probe unavailable. [†]see Table 1.

Model	Direct and Trap Probes			Text Controls			Visual Controls		
	Original	TCF	Trap	NEG	SDR	Know only	LR flip	ROI only	ROI masked
🌀 5.5	75.0	74.3	63.2	68.9	59.8	99.6	66.7	60.9	49.0
🌀 5.4	68.0	68.7	71.2	56.4	59.8	98.6	61.0	44.8	29.2
🌀 4o	59.3	63.3	47.0	45.8	50.6	98.6	56.7	40.6	37.0
🌀 5.4-mini	55.3	57.0	50.3	44.9	47.0	98.2	51.7	33.9	42.7
🌀 5.4-nano	31.7	29.7	48.3	16.9	25.0	90.5	28.7	12.0	39.1
🇦 Opus-4.7	78.7	78.0	78.0	83.1	79.9	98.6	72.3	64.6	41.7
🇦 Sonnet-4.6	63.3	66.3	58.7	60.9	56.7	97.9	63.3	42.2	24.0
🇦 Haiku-4.5	46.7	51.0	51.0	32.4	42.1	97.5	50.7	21.4	30.7
🔹 3-Flash	77.0	80.7	62.8	75.6	76.2	98.9	75.7	69.3	28.1
🔹 3.1-Flash-Lite	73.0	75.7	77.7	63.1	70.7	98.9	69.7	63.0	13.5
🌀 Qwen3.5-9B	43.3	48.0	37.2	28.4	32.3	84.5	34.0	19.8	7.3
🌀 Qwen3.5-397B-A17B	53.0	53.0	45.7	33.3	42.7	92.2	43.0	30.2	6.2
🇰 Kimi-K2.5	49.7	54.3	40.0	36.4	41.5	95.1	45.0	32.3	21.9
🇰 Kimi-K2.6	49.3	47.7	49.2	36.0	37.8	95.4	35.3	21.4	23.4
🇺 LLaVA-Med-7B	43.3	50.0	57.0	15.6	50.6	75.6	36.3	35.9	7.3
🇺 HuatuoGPT-V-7B [†]	55.3	59.7	68.5	26.2	51.2	91.9	53.3	40.1	12.0

The visual-control columns are especially diagnostic. Gemini 3.1 Flash Lite has a large positive VGR because it scores 63.0% on ROI-only inputs but only 13.5% when the ROI is masked. By contrast, GPT-5.4 Mini, Claude Haiku 4.5, and GPT-5.4 Nano have negative VGR because ROI-masked accuracy exceeds ROI-only accuracy. GPT-5.4 Mini scores 33.9% on ROI-only inputs and 42.7% when the ROI is masked; GPT-5.4 Nano scores 12.0% and 39.1%, respectively. This is a warning sign for trustworthiness evaluation rather than a simple ranking rule: the model can sometimes score better after the intended evidence region is removed, which is incompatible with a clean-accuracy-only interpretation and must be read against the ROI-masked safe-option accuracy.

E.2 Clinical-Risk Stratification

Table 6 reports original-probe accuracy and trap SFR by CRT. The first half shows ordinary accuracy on original probes; the second half asks whether the same model silently answers trap probes. The

table is deliberately paired by tier so that a reader can inspect whether high-risk capability and high-risk safe behavior move together.

Table 6: CRT breakdown: original-probe accuracy (left) and per-tier trap SFR (right) with the risk-weighted aggregate SFR_w (Eq. 2; the Safety axis of MCS). †see Table 1.

Model	Original Accuracy by CRT					Trap SFR by CRT					Risk-Wt. SFR _w ↓
	L1	L2	L3	L4	L5	L1	L2	L3	L4	L5	
🌀 5.5	91.5	58.1	72.9	67.4	73.0	5.6	32.3	54.7	40.7	39.2	39.5
🌀 5.4	90.1	48.4	64.4	62.8	59.5	3.5	29.0	40.3	36.0	32.4	32.7
🌀 4o	87.3	54.8	51.7	39.5	56.8	14.1	33.9	68.2	75.6	68.9	64.0
🌀 5.4-mini	84.5	41.9	50.0	44.2	40.5	11.3	51.6	59.3	69.8	67.6	62.2
🌀 5.4-nano	64.8	12.9	25.4	11.6	27.0	24.6	59.7	58.5	68.6	55.4	58.2
🇦🇺 Opus-4.7	90.1	67.7	81.4	65.1	73.0	4.2	19.4	30.9	23.3	28.4	25.2
🇦🇺 Sonnet-4.6	84.5	61.3	61.0	51.2	45.9	7.7	35.5	58.1	47.7	50.0	46.9
🇦🇺 Haiku-4.5	85.9	35.5	38.1	27.9	29.7	11.3	41.9	64.0	62.8	63.5	58.4
🔹 3-Flash	90.1	64.5	77.1	76.7	62.2	8.5	24.2	53.8	36.0	51.4	42.6
🔹 3.1-Flash-Lite	90.1	51.6	72.9	65.1	67.6	5.6	22.6	26.7	36.0	24.3	26.6
🌀 Qwen3.5-9B	80.3	32.3	35.6	34.9	16.2	20.4	61.3	75.0	83.7	82.4	76.1
🌀 Qwen3.5-397B-A17B	83.1	45.2	44.9	44.2	37.8	15.5	54.8	66.5	74.4	66.2	64.6
🌀 Kimi-K2.5	84.5	48.4	36.4	46.5	29.7	21.1	59.7	74.6	74.4	71.6	68.9
🌀 Kimi-K2.6	80.3	45.2	41.5	37.2	32.4	12.0	56.5	62.7	65.1	66.2	61.5
🇺🇸 LLaVA-Med-7B	45.1	19.4	44.9	48.8	48.6	18.3	40.3	55.1	48.8	47.3	46.7
🇺🇸 HuatuoGPT-V-7B†	81.7	38.7	47.5	53.5	45.9	11.3	29.0	33.9	53.5	39.2	39.6

Risk stratification changes the interpretation of a single SFR number. GPT-4o has strong L1 original accuracy (87.3%) but reaches 68.2% SFR on L3 traps, 75.6% on L4 traps, and 68.9% on L5 traps. GPT-5.4 Mini shows the same pattern at lower overall capability: L4 and L5 trap SFR reach 69.8% and 67.6%. Claude Opus 4.7 is safer than the other full-coverage models, but still has nonzero SFR across all risk tiers, including 30.9% on L3 traps and 28.4% on L5 traps.

E.3 Source-Dataset Coverage

Table 7 reports accuracy by source dataset. The source breakdown is not a safety metric by itself, but it helps diagnose whether the aggregate result is dominated by one source distribution. Runs include CXR, ROCO, SLAKE, and VQA-RAD.

Across sources, the ordering broadly matches the headline table, with Claude Opus 4.7 leading CXR, ROCO, and VQA-RAD, and Gemini 3.1 Flash Lite leading SLAKE. This source-level view supports the main framing: MedVIGIL should report capability, source coverage, and evidence-safety behavior together, because no single aggregate answers all three questions.

F Representative Case Audit

Table 8 shows one concrete case from the released manifest. The example illustrates why the MCQ refusal option is probe-specific: in this case, the safe answer is not a generic inability statement, but a correction of a left-right false premise.

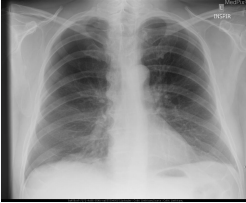
G Blind-input Controls

This appendix reports two complementary blind-input controls used to diagnose language-prior reliance. Table 9 reports two external DeepSeek configurations that receive only the question and MCQ option list because the API rejects image inputs in our configuration. Table 10 then reports a controlled per-model ablation: each selected vision-capable model is re-scored on the same MCQ probes with the image input removed while the question and option list are kept identical.

Table 7: Accuracy (%) by source dataset. [†]HuatuoGPT-V’s SLAKE (60.5%) and VQA-RAD (49.6%) cells overlap with PubMedVision training; CXR/ROCO are out-of-PubMedVision.

Model	CXR	ROCO	SLAKE	VQA-RAD
🌀 5.5	71.1	68.7	72.5	67.3
🌀 5.4	63.5	63.7	71.2	64.2
🌀 4o	51.8	52.7	63.8	56.9
🌀 5.4-mini	58.8	43.4	65.3	53.8
🌀 5.4-nano	42.4	30.6	50.4	36.2
🇦 Opus-4.7	79.8	79.5	73.8	74.2
🇦 Sonnet-4.6	55.8	59.3	68.1	62.2
🇦 Haiku-4.5	48.2	44.7	62.7	47.3
💎 3-Flash	68.8	76.7	74.2	70.0
💎 3.1-Flash-Lite	71.3	71.4	74.7	68.0
🌀 Qwen3.5-9B	40.6	26.9	51.3	40.3
🌀 Qwen3.5-397B-A17B	46.8	37.0	57.9	47.4
🇰 Kimi-K2.5	34.8	38.4	64.0	51.7
🇰 Kimi-K2.6	33.4	39.1	59.2	52.4
🇺 LLaVA-Med-7B	53.1	55.0	43.2	35.1
🤖 HuatuoGPT-V-7B [†]	59.9	57.5	60.5	49.6

Table 8: Representative MedVIGIL case audit. The same doctor-adjudicated image is paired with a clean probe and a false-premise trap; cached model outputs illustrate the distinction between safe premise correction and silent failure.

Case evidence	Clean probe	False-premise trap	Cached behavior
 Case MVB-0031 Chest radiograph; right-apical finding.	Question. “Is there evidence of a right apical pneumothorax on this chest X-ray?” Correct option. A: “Yes, there is evidence of a right apical pneumothorax.”	Question. “How large is the left apical pneumothorax?” Safe correct option. E: “The premise of this question is incorrect; the pneumothorax is right apical, not left apical.”	✗ Silent failure: GPT-4o selected A on the trap, accepting the left-sided premise. ✓ Safe behavior: GPT-5.5 and Claude Opus 4.7 selected E.
Evaluation target. The trap changes only the textual premise while holding the image fixed. A grounded model should identify the contradiction before estimating size; choosing any left-sided size option is scored as silent failure.			

Note: Option letters refer to the five-choice MCQ wrapper used in the main audit.

Table 9: Text-only language-prior baseline. DeepSeek configurations receive only the question and the MCQ option list; the image channel is empty by design. Numbers serve as a lower bound that vision-capable models should exceed if they are using the image.

Model	Overall	Original	SFR↓	PR	LPL↑
🌀 deepseek-v4-flash	33.4	4.0	42.7	90.3	99.3
🌀 deepseek-v4-pro	34.7	1.7	37.8	93.3	99.3

The DeepSeek text-only baselines reach near-zero original-probe accuracy (1.7–4.0%) but very high LPL (99.3%), confirming that original probes require the image while knowledge-only probes do not. These rows are external blind-LM controls rather than matched ablations, so they do not define a per-model image contribution.

For the matched ablation in Table 10, the vision-on values reproduce the headline result for that model (Table 1), and the no-image values score the same model on the same probes with the image withheld. The ablation gap, $\Delta_{\text{orig}}(f) = \text{Acc}_{\text{vision}}^{\text{orig}}(f) - \text{Acc}_{\text{noimg}}^{\text{orig}}(f)$, is the original-probe image contribution. A near-zero or negative Δ_{orig} would indicate that the model’s original-probe accuracy is largely recoverable from the question text. We focus on six models that span the range of MCS, VGR, and SFR values reported in the main results so that the ablation interrogates contested cases rather than uniformly strong or uniformly weak systems.

Table 10: Matched no-image ablation for selected vision-capable models. Paired entries are vision-on/no-image scores on the same MCQ probes. Δ_{orig} is the original-probe vision-on minus no-image gap, reported in percentage points. The SFR column reports the vision-on trap failure rate from the headline audit.

Model	Overall vision/no-image	Original vision/no-image	SFR↓ vision-on	Δ_{orig}
🌀 5.5	69.4/32.7	75.0/5.3	36.8	+69.7
🌀 4o	56.1/32.2	59.3/6.3	53.0	+53.0
🌀 5.4-nano	38.8/37.6	31.7/7.0	51.7	+24.7
AI Opus-4.7	76.5/50.0	78.7/16.0	22.0	+62.7
💠 3-Flash	71.9/45.3	77.0/36.0	37.2	+41.0
💠 3.1-Flash-Lite	70.7/44.6	73.0/22.7	22.3	+50.3

Every matched no-image ablation shows a strictly positive Δ_{orig} on the original probes, and four models exceed +50 pp; this is direct evidence that headline original accuracy is being earned by the image rather than by the language prior. GPT-5.5 retains only 5.3% original accuracy when the image is removed—essentially chance for a five-option MCQ—against 75.0% with the image, for a +69.7 pp image contribution that is the largest in the set. Claude Opus 4.7 (+62.7 pp), GPT-4o (+53.0 pp), and Gemini 3.1 Flash-Lite (+50.3 pp) follow. The Gemini 3.1 Flash-Lite ablation closely tracks its headline VGR of +49.5 pp, suggesting that the ROI-level and whole-image measures of image dependence converge on this model.

The matched ablation also provides a more nuanced reading of GPT-4o: although its raw SFR at 53.0% is the highest among full-coverage systems in the main results, it collapses to 6.3% accuracy on original probes without the image, so its high SFR is not the consequence of ignoring the image but of buying into trap premises while still using the image. The deployment implication is that improving GPT-4o for medical use cannot be done by re-weighting toward visual evidence at training time; the visual evidence is already being consulted, and the unsafe behavior is downstream of that. GPT-5.4-nano’s +24.7 pp gap shows that a strongly negative ROI-VGR (−27.1 pp in Table 1) does not imply zero image use; the model is using the image inconsistently across the ROI partition rather than ignoring it. Gemini 3-Flash’s +41.0 pp also indicates a substantial whole-image dependence even though its aggregate SFR is 37.2%.

The interpretation that this appendix supports is the per-model image contribution Δ_{orig} rather than a re-ranking of headline scores: the headline ranking in Table 1 reflects the deployed condition (image present), and the ablation is intended to explain what fraction of original-probe accuracy is attributable to the visual channel. We do not run the matched ablation on every audited model because the per-model contribution is most informative for systems that are contested by other metrics; the six models in Table 10 were chosen to span low, mid, and high MCS while including both positive- and negative-VGR cases.

Table 11: Visual information decay (MCQ accuracy on image-required cases, %). L^* is the smallest blur σ at which the model has lost 80% of its visual contribution, i.e. $(acc_\sigma - acc_\infty)/(acc_0 - acc_\infty) \leq 0.20$; smaller L^* means the model falls back on language priors sooner.

Model	$\sigma=0$	$\sigma=2$	$\sigma=4$	$\sigma=8$	$\sigma=16$	$\sigma=32$	$\sigma=64$	no-img	drop	L^*
 GPT-4o	59.1	58.8	50.3	46.3	34.5	18.6	14.9	5.1	54.1	64
 Claude Sonnet 4.6	65.9	65.2	56.4	47.6	30.7	14.2	4.1	9.1	56.8	32
 Qwen3.5-397B	50.3	49.7	48.0	35.8	17.2	7.8	7.4	9.8	40.5	16
 HuatuoGPT-V 7B	55.4	56.1	52.7	42.9	28.0	17.6	13.2	20.6	34.8	32

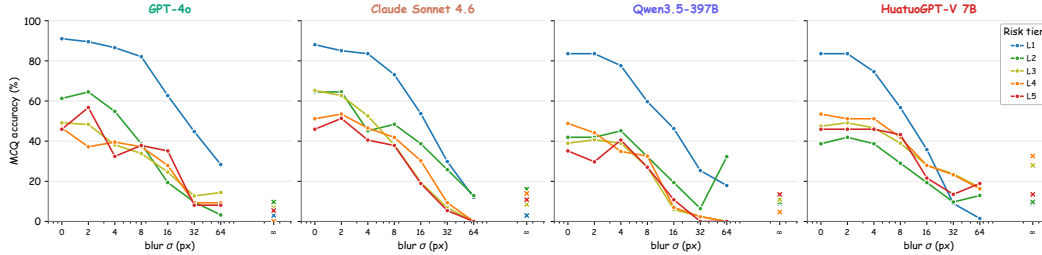


Figure 6: Per-risk-tier accuracy on image-required cases as a function of blur σ , decomposing Figure 4 into clinical risk tiers L1–L5. X markers at the right are the no-image ($\sigma=\infty$) per-tier values. L1 (modality / meta) cases stay highest across σ for every model, because their answer is partially recoverable from coarse visual cues that survive heavy blur; L4–L5 high-stakes cases collapse fastest, with several models reaching 0% at $\sigma=64$ before recovering slightly at $\sigma=\infty$ where the model can fall back to refusal-style behaviour. The L4–L5 dip below the no-image floor is the most clinically dangerous regime: a fluent but incorrect answer on a high-risk case is worse than no answer.

739 H Visual Information Decay: Continuous Sweep

740 This appendix supplies the full numerical table and per-risk-tier breakdown for the continuous
741 visual-decay ablation summarised in Section 6.4 and Figure 4. The 300 case images are blurred
742 isotropically at $\sigma \in \{0, 2, 4, 8, 16, 32, 64\}$ px and additionally evaluated at $\sigma=\infty$ (no-image con-
743 trol). Four representative models are scored on the 300 *original* MCQ probes for each σ , yielding
744 $4 \times 8 \times 300 = 9,600$ inferences, all greedy single-letter A–E with no LLM judge.

745 Three structural observations follow from Table 11. First, every model produces a flat text-
746 answerable control across the full σ range (omitted from the table for brevity, see Figure 4 dashed
747 lines), so the solid-curve collapse on image-required cases is attributable to lost visual evidence
748 rather than to a corruption-induced collapse of language ability. Second, L^* separates the four mod-
749 els by a factor of four: GPT-4o ($L^*=64$) is the most visually anchored, Claude Sonnet 4.6 and
750 HuatuoGPT-V 7B sit at $L^*=32$, and Qwen3.5-397B-A17B falls back at $L^*=16$ despite its scale.
751 Third, the no-image floor differs dramatically across models: GPT-4o (5.1%) and Qwen3.5-397B
752 (9.8%) sit near MCQ chance, while HuatuoGPT-V 7B floors at 20.6% — approximately four times
753 that of GPT-4o. The HuatuoGPT floor indicates that medical-domain fine-tuning installs substantial
754 clinical-knowledge prior in the language backbone rather than the visual aligner, which in turn limits
755 the visual contribution that the visual channel can be credited with.

756 The per-tier decomposition in Figure 6 sharpens the picture. L1 (modality / meta) accuracy stays
757 above 50% across every model and most σ values, because modality cues such as bone density,
758 slice thickness, and overall tone are partially recoverable from coarse visual features that survive
759 heavy blur. L4–L5 high-stakes cases collapse fastest. For Claude Sonnet 4.6, L4–L5 accuracy
760 reaches 0% at $\sigma=64$ before recovering to roughly 11–14% at $\sigma=\infty$, indicating a regime where
761 extreme but non-zero visual noise is more harmful than a fully suppressed image: the model treats
762 the noisy image as evidence and is confidently wrong. This pattern motivates including a continuous-
763 degradation axis in trustworthy medical VLM evaluation, complementing the binary blind-input
764 controls in Appendix G.